The χ^2 Test as a Test of Independence

With the help of the χ^2 test, we can find whether or not two attributes are associated. We take the null hypothesis that there is no association between the attributes under study, *i.e.*, **we assume that the two attributes are independent**. **If the calculated value of χ^2 is less than the table value** at a specified level (generally 5%) of significance, the hypothesis holds true, *i.e.*, **the attributes are independent** and do not bear any association. On the other hand, if the calculated value of χ^2 is greater than the table value at a specified level of significance, we say that the results of the experiment do not support the hypothesis. In other words, the attributes are associated. Thus a very useful application of the χ^2 test is to investigate the relationship between trials or attributes, which can be classified into two or more categories.

A				B			
	B_1	B_2	-	B_j	-	B_n	Row Total
A_1	O_{11}	O_{12}	-	O_{1j}	-	O_{1n}	O_{1*}
A_2	O_{21}	O_{22}	-	O_{2j}	-	O_{2n}	O_{2*}
:	-	-	-	-	-	-	-
A_i	O_{i1}	O_{i2}	-	O_{ij}	-	O_{in}	O_{i*}
:	-	-	-	-	-	-	-
A_m	O_{m1}	O_{m2}		O_{mj}	-	O_{mn}	O_{m*}
Column Total	O_{*1}	O_{*2}	-	O_{*j}	-	O_{*n}	N

Now, based on the null hypothesis H_0, i.e. the assumption that the two attributes A and B are independent, we compute the expected frequencies E_{ij} for various cells, using the following formula:

$$E_{ij} = \frac{O_{i*} \times O_{*j}}{N}, \; i = 1, 2, \ldots, m; \quad j = 1, 2, \ldots, n$$

i.e. $$\boxed{E_{ij} = \left(\frac{\text{Total of observed frequencies in the } ith \text{ row} \times \text{ total of a observed frequencies in the } j\text{th column}}{\text{Total of all cell frequencies}} \right)}$$

Then we compute

$$\boxed{\chi^2 = \sum_{i=1}^{m} \sum_{j=1}^{n} \left\{ \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \right\}}$$

The number of degrees of freedom for this χ^2 computed from the $(m \times n)$ contingency table is $\boxed{\nu = (m - 1)(n - 1).}$

If $\chi^2 < \chi^2_\nu(\alpha)$, H_0 is accepted at α % LOS, i.e. the attributes A and B are independent.

If $\chi^2 > \chi^2_\nu(\alpha)$, H_0 is rejected at α % LOS, i.e. A and B are not independent.

What are the expected frequencies of the 2 × 2 contingency tables given below:

(i)

a	b
c	d

(ii)

2	10
6	6

Soln:-

Observed frequency

(i)

$a^{O_{11}}$	$b^{O_{12}}$	$a+b$
$c^{O_{21}}$	$d^{O_{22}}$	$c+d$
$a+c$	$b+d$	$a+b+c+d$

Expected Frequency

E_{11} $\frac{(a+b)(a+c)}{a+b+c+d}$	$\frac{(a+b)(b+d)}{a+b+c+d}$ E_{12}
$\frac{(c+d)(a+c)}{a+b+c+d}$ E_{21}	$\frac{(b+d)(c+d)}{a+b+c+d}$ E_{22}

(ii)

$2^{O_{11}}$	$10^{O_{12}}$	12
O_{21} 6	$6^{O_{22}}$	12
8	16	24

$\frac{12 \times 8}{24} = 4$ E_{11}	$\frac{12 \times 16}{24} = 8$ E_{12}
$\frac{12 \times 8}{24} = 4$ E_{21}	$\frac{12 \times 16}{24} = 8$ E_{22}

From the following table regarding the color of eyes of fathers and sons test whether the color of the son's eye is associated with that of the father.

		Eye color of son	
		Light	*Not light*
Eye color of father	*Light*	*471*	*51*
	Not light	*148*	*230*

Soln:- H_0: The color of the son's eye is not associated with that of father

H_1: The color of the son's eye is associated with that of father.

Observed Frequency		Row sum
$O_{11} = 471$	$O_{12} = 51$	522
$O_{21} = 148$	$O_{22} = 230$	378
Column Sum 619	281	900

Expected Frequency

$E_{11} = \frac{522 \times 619}{900}$	$E_{12} = \frac{522 \times 281}{900}$
$E_{21} = \frac{378 \times 619}{900}$	$E_{22} = \frac{378 \times 281}{900}$

$E_{11} = 359.02$	$E_{12} = 162.98$
$E_{21} = 259.08$	$E_{22} = 118.02$

Now, $$\chi^2 = \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= \frac{(O_{11} - E_{11})^2}{E_{11}} + \frac{(O_{12} - E_{12})^2}{E_{12}} + \frac{(O_{21} - E_{21})^2}{E_{21}} + \frac{(O_{22} - E_{22})^2}{E_{22}}$$

$$= \frac{(471 - 359.02)^2}{359.02} + \frac{(51 - 162.98)^2}{162.98} + \frac{(148 - 259.08)^2}{259.08} + \frac{(230 - 118.02)^2}{118.02}$$

$$= 265.74$$

$1 = (2-1)(2-1)$

$\chi^2_{0.05}$ for 1 degrees of freedom is 3.841

Since $\chi^2 > \chi^2_{0.05}$, we reject the null hypothesis.

The following table gives the number of good and bad parts produced by each of the three shifts in a factory:

	Good parts	Bad parts	Total
Day shift	960	40	1000
Evening shift	940	50	990
Night shift	950	45	995
Total	2850	135	2985

Test whether or not the production of bad parts is independent of the shift on which they were produced.

Soln: –

H_0 : The production of bad parts is independent of the shift on which day they were produced.

H_1 : The production of bad part is not independent of the shift on which day they were produced.

Observed	Frequency	Rowsum
$O_{11} = 960$	$O_{12} = 40$	1000
$O_{21} = 940$	$O_{22} = 50$	990
$O_{31} = 950$	$O_{32} = 45$	995
Column sum 2850	135	2985

Expected Frequency

$E_{11} = \frac{1000 \times 2850}{2985} = 954.77$ $E_{12} = \frac{1000 \times 135}{2985} = 45.27$

$E_{21} = \frac{990 \times 2850}{2985} = 945.226$ $E_{22} = \frac{990 \times 135}{2985} = 44.773$

$E_{31} = \frac{995 \times 2850}{2985} = 950$ $E_{32} = \frac{995 \times 135}{2985} = 45$

$$\chi^2 = \frac{(O_{11} - E_{11})^2}{E_{11}} + \frac{(O_{12} - E_{12})^2}{E_{12}} + \frac{(O_{21} - E_{21})^2}{E_{21}} + \frac{(O_{22} - E_{22})^2}{E_{22}} + \frac{(O_{31} - E_{31})^2}{E_{31}} + \frac{(O_{32} - E_{32})^2}{E_{32}}$$

$= 1.28126$

$\chi^2_{0.05}$ for degrees of freedom $(3-1)(2-1) = 2$ is 5.991

Since $\chi^2 < \chi^2_{0.05}$ we accept the null hypothesis.

From the following data, find whether hair color and sex are associated.

Sex \ Color	Fair	Red	Medium	Dark	Black	Total
Boys	592	849	504	119	36	2100
Girls	544	677	451	97	14	1783
Total	1136	1526	955	216	50	3883